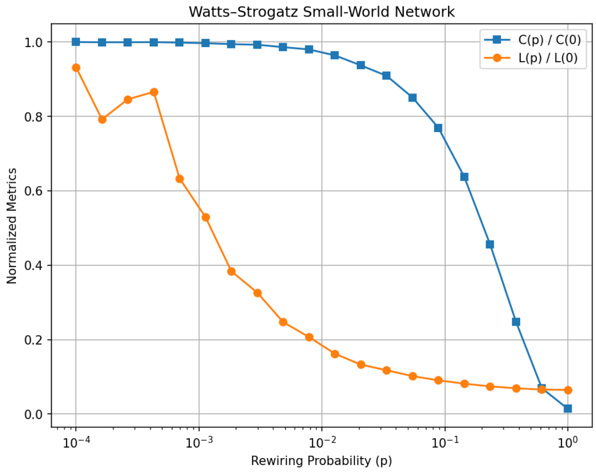
CSE 655 — Network Science Assignment 2 Report

Winter 2026

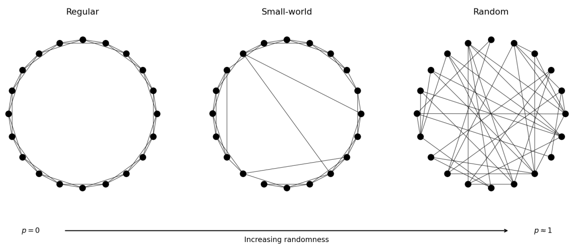
Q1 WATTSSTROGATZ SMALL-WORLD MODEL

Parameters: $n = 1000$, $k = 10$, $p \in [10^{-4}, 1]$ (log scale, 20 points, 10 runs each).

We start with a ring lattice and rewire each edge with probability p. Metrics are normalised by their $p = 0$ values: $C(0) = 0.667$, $L(0) = 50.45$.



(a) Scaled $C(p)/C(0)$ and $L(p)/L(0)$ vs rewiring probability.



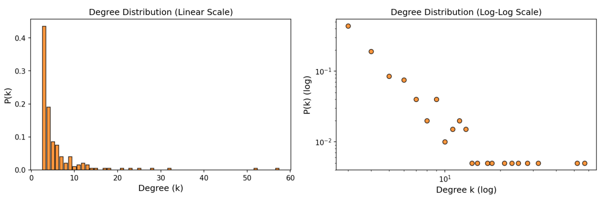
(b) Network structure at $p = 0$, $p \approx 0.15$, $p = 1$.

Path length collapses quickly — even rewiring a tiny fraction of edges is enough to create shortcuts across the network. Clustering stays high for longer because most edges are still local. The interesting range is around $p = 0.01$–0.1, where both properties hold at the same time. This is the small-world regime.

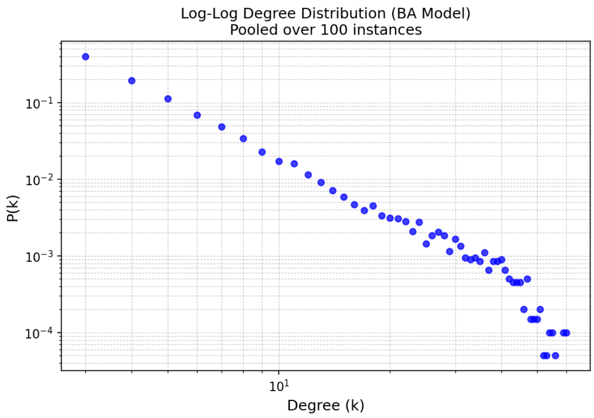
Q2 BARABÁSI-ALBERT SCALE-FREE NETWORK

Parameters: $n = 200$, $m_0 = 5$, $m = 3$. The BA algorithm was implemented from scratch using preferential attachment $P_i \propto k_i / \sum_j k_j$.

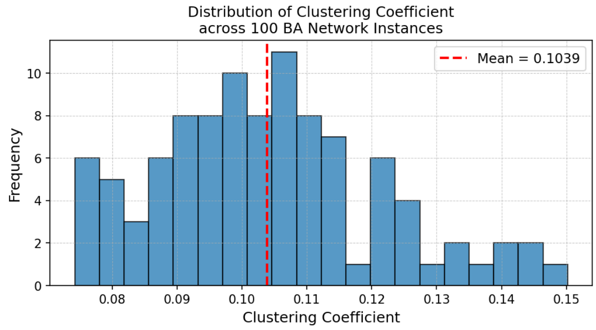
Note on rubric: "a minimum of 100 instances" is ambiguous — we address both interpretations: a single network with 200 nodes, and an average over 100 independently generated networks.



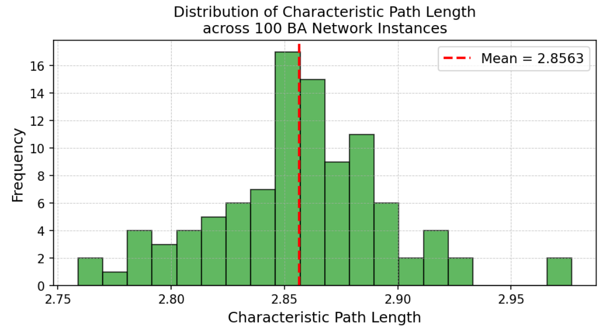
(a) Degree distribution of a single BA network ($n = 200$).



(b) Log-log degree distribution pooled over 100 instances.



(a) Distribution of average clustering coefficient over 100 instances.



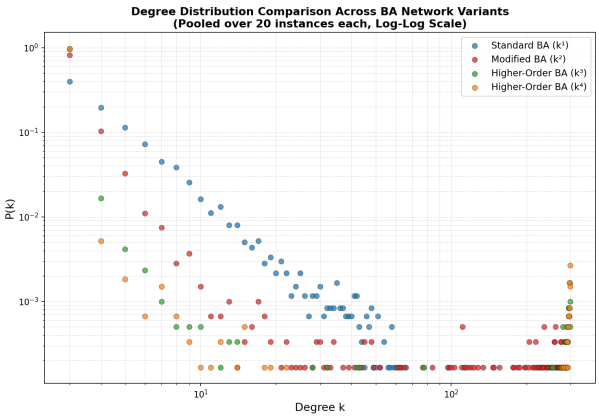
(b) Distribution of characteristic path length over 100 instances.

Over 100 instances: $\langle C \rangle = 0.105 \pm 0.018$, $\langle L \rangle = 2.855 \pm 0.038$. The log-log plot is roughly a straight line, consistent with a power law ($P(k) \propto k^{-3}$). Clustering is low because new nodes pick targets based on degree rather than proximity, so local triangles rarely form. Despite that, paths are short because hubs link distant parts of the network.

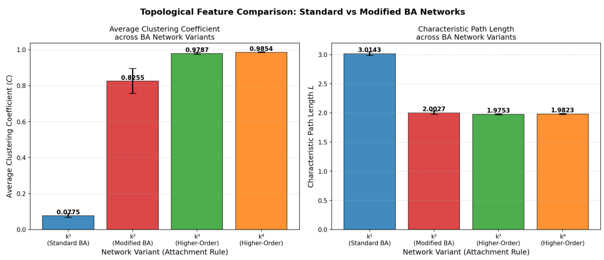
Q3 MODIFIED BA: K^α PREFERENTIAL ATTACHMENT

Parameters: $n = 300$, $m_0 = 5$, $m = 3$, 20 instances per variant, $\alpha \in \{1, 2, 3, 4\}$.

Attachment probability: $\Pi(k_i) \propto k_i^\alpha / \sum_j k_j^\alpha$. At $\alpha = 1$ this is standard BA. Increasing α makes the rich-get-richer effect stronger — the highest-degree nodes capture an even bigger share of incoming connections.



(a) Degree distributions for all four α variants (log-log).



(b) Average CC and path length across variants (error bars = std).

Variant	Avg $\langle C \rangle$	Std	Avg L	Std
Standard BA (k^1)	0.078	0.011	3.014	0.031
Modified BA (k^2)	0.826	0.069	2.003	0.028
Higher-order (k^3)	0.979	0.006	1.975	0.011
Higher-order (k^4)	0.985	0.002	1.982	0.010

Table 1: Topological metrics averaged over 20 instances per variant.

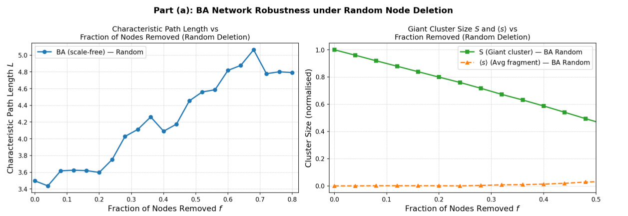
One result that caught us off guard: clustering goes *up* with α, not down. The reason is that with higher α, all new nodes end up connecting to the same small group of initial hub nodes, which are already mutually connected (they form the starting clique). So almost every new node's neighbours are all connected to each other, giving high clustering. Path length falls from around 3 at $\alpha = 1$ to around 2 at $\alpha = 2$ and barely changes after that — once everything routes through one hub, distances can't get much shorter.

Q4 NETWORK ROBUSTNESS: NODE DELETION

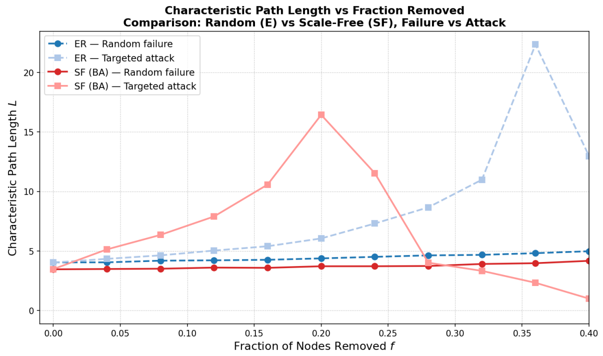
Two strategies: *random failure* (remove a random node) and *targeted attack* (always remove the highest-degree node remaining). We track S (giant cluster size / n_0), $\langle s \rangle$ (mean non-giant fragment size / n_0), and path length L.

(a) & (b) — BA vs ER Networks

BA network: $n = 1000$, $m = 3$, avg degree 5.98. ER network: same n, edge probability matched to give avg degree 6.02.



(a) $L(f)$ and $S(f)$ under random deletion on the BA network.



(b) Path length $L(f)$ — ER vs BA, random failure vs targeted attack.

For the ER network, both strategies look pretty similar — there are no special nodes to target, so it doesn't matter much which ones you remove. The BA network tells a different story: random removal barely dents it (most nodes are low-degree), but removing just the top hubs causes rapid collapse. This asymmetry is exactly what Albert et al. (2000) describe.

(c) — Real-World Network: email-Eu-core

We used the **email-Eu-core** dataset from the Stanford Network Analysis Project (SNAP) [1], which captures email communication between members of a European research institution. The dataset has 1005 nodes and 25571 edges (avg degree 50.88), fits within the required $500 < n < 5000$ range, and has a heavy-tailed degree distribution (max degree 347) confirming scale-free structure.

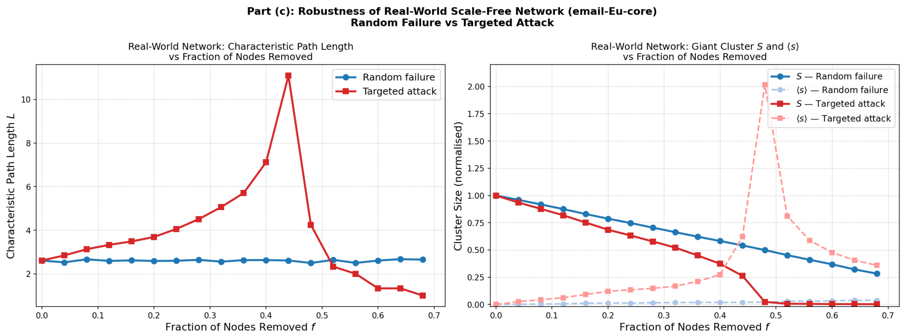


Figure 6: S and $\langle s \rangle$ on email-Eu-core under random failure and targeted attack.

The email network holds up well under random node removal — S drops slowly and the network stays largely connected. Under targeted attack it fragments much faster; removing even the top 5% of nodes (the heaviest emailers) has a noticeable effect.

(d) — Comparison with Albert et al. (2000)

Albert et al. found that ER networks have a percolation threshold around $f_c \approx 0.28$ regardless of strategy, while scale-free networks have no real threshold under random failure but collapse at $f_c \approx 0.18$ under targeted attack. Our results follow the same pattern. The curves are a bit noisier than in the paper because our networks are smaller ($n \sim 1000$ vs ~ 10000) and we sample path lengths rather than compute them exactly, but the core finding holds.

Q5 YEAST PROTEIN INTERACTOME (YPI)

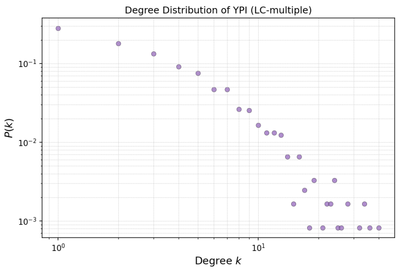
We used two external datasets:

- **LC-multiple interaction data** from the CCSB Yeast Interactome Database [2] (`http://interactome.dfci.harvard.edu/S_cerevisiae/`), containing literature-curated protein–protein interactions for *S.*

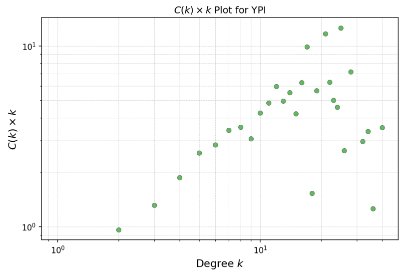
cerevisiae. After loading: 1213 proteins, 2556 interactions, average degree 4.21, max degree 40.

- **Essential gene annotations** from the Saccharomyces Genome Database (SGD) [3] (https://www.yeastgenome.org/). A gene is labelled essential if its null mutant has an "inviable" phenotype in the SGD phenotype data file.

(a) — Network Properties



(a) Degree distribution (log-log).



(b) $C(k) \times k$ plot.

(c) Cytoscape visualisation (node size $\propto$ degree).

The degree distribution has a heavy tail — most proteins have just one or two interactions, while a handful of hubs have many. The $C(k) \times k$ plot shows that low-degree proteins tend to sit in tight clusters, while high-degree hubs have lower local clustering, suggesting they connect different parts of the network rather than participating in one dense group.

(b) — Essential Proteins

568 of 1213 proteins (46.8%) are essential according to the SGD data. Among the most connected proteins the fraction is much higher — the top hub (YDL140C, degree 40) is essential, and so are most of the next highest-degree proteins.

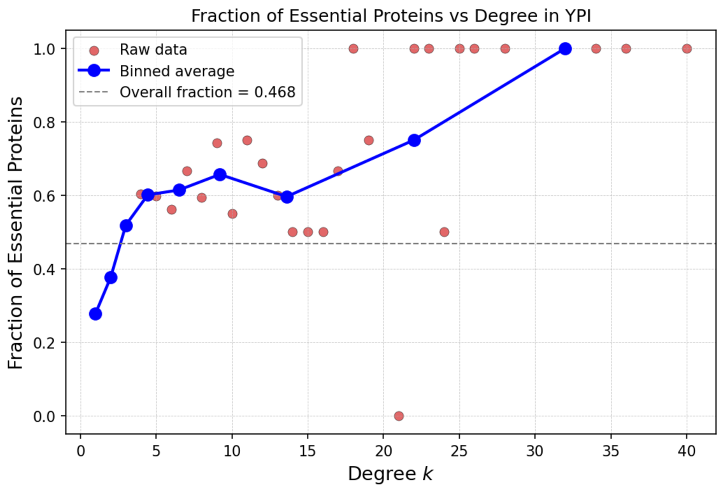


Figure 8: Fraction of essential proteins vs degree k. The trend is clear above $k \approx 5$.

This is the **centrality-lethality rule** (Jeong et al., *Nature* 2001): proteins that interact with many others are harder to remove without breaking something important. A low-degree protein can usually be compensated for; a hub cannot.

(c) — Robustness of YPI

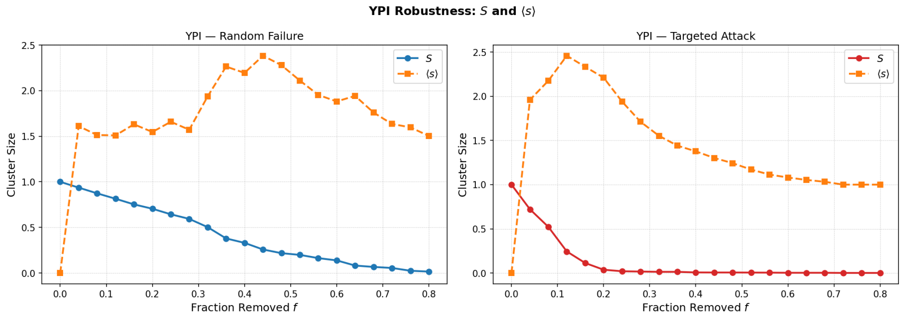


Figure 9: S and $\langle s \rangle$ under random failure and targeted attack on YPI.

The YPI behaves like the other scale-free networks we tested: random removal has little effect, but taking out the hubs causes the network to break apart quickly. What makes Q5 interesting is that the proteins removed first in a targeted attack are exactly the ones identified as essential in Part (b). The structural importance and biological importance of these proteins line up.

CONCLUSIONS

1. **Q1 — WS model:** The small-world regime sits in a narrow window around $p = 0.01$–0.1. Outside it, the network is either too regular or too random to have both properties at once.
2. **Q2/Q3 — BA model:** Preferential attachment gives a power-law degree distribution. Raising the attachment exponent α concentrates connections further, and — perhaps surprisingly — increases clustering because new nodes all end up attached to the same initial hub group.
3. **Q4 — Robustness:** Scale-free networks handle random failures well but fall apart under targeted attacks on hubs. ER networks don't show this split. The email-Eu-core network follows the same pattern as the synthetic BA network.
4. **Q5 — YPI:** The yeast protein network is scale-free, and its hubs are disproportionately essential genes. Structural centrality and biological essentiality point to the same proteins.

Data sources. Q4: email-Eu-core network from SNAP [1] (https://snap.stanford.edu/data/email-Eu-core.html). Q5: LC-multiple protein interactions from CCSB Yeast Interactome Database [2] (http://interactome.dfci.harvard.edu/S_cerevisiae/); essential gene phenotypes from SGD [3] (https://www.yeastgenome.org/).

Note. Some graphs may not be included in this report due to page limit constraints. All corresponding plots and visualizations are provided in the submission folders for each respective question. Also all the results are there in the notebooks even if they might have been missed from being mentioned in the report.

References

[1] J. Leskovec and A. Krevl, *SNAP Datasets: Stanford Large Network Dataset Collection*, http://snap.stanford.edu/data, 2014.

[2] H. Yu et al., "High-quality binary protein interaction map of the yeast interactome network," *Science*, 322(5898):104–110, 2008.

[3] SGD Project, *Saccharomyces Genome Database*, https://www.yeastgenome.org/.

[4] R. Albert, H. Jeong, A.-L. Barabási, "Error and attack tolerance of complex networks," *Nature*, 406:378–382, 2000.

[5] H. Jeong et al., "Lethality and centrality in protein networks," *Nature*, 411:41–42, 2001.